

Frozen Bubble Flake Display: A Novel Projection Screen using a Frozen Soap Bubble

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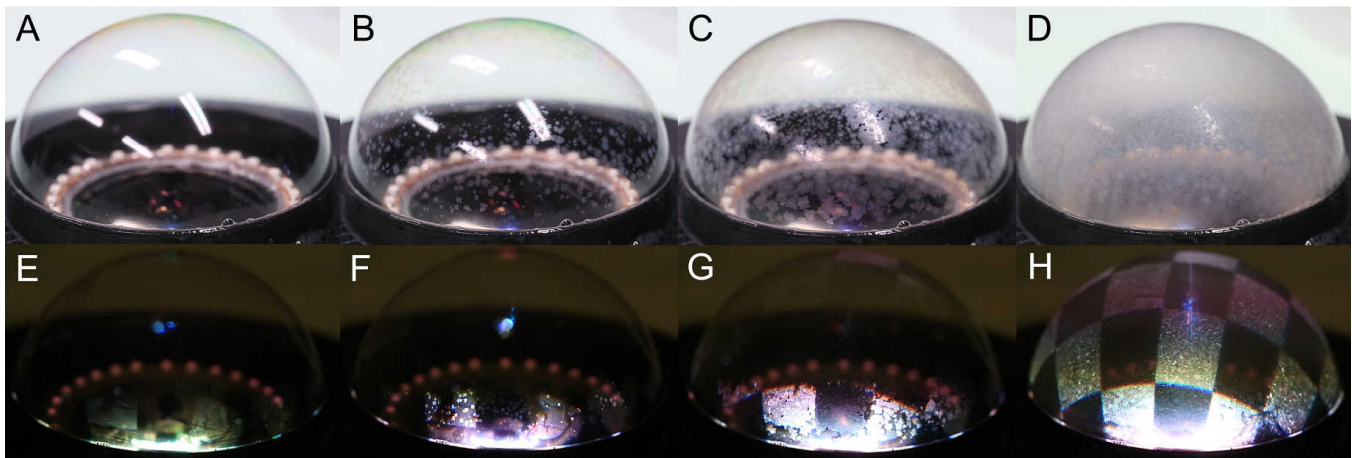


Figure 1: Figure of the freezing process step by step. Freezing process of soap bubble (A-D) and projection onto frozen soap bubble (E-H).

Abstract

We propose a novel projection screen material that utilizes the freezing phenomenon of soap film. Soap film is naturally transparent, but when frozen at low temperatures, it becomes more opaque and can serve as a projection screen. In this research, we focus on the frozen soap film and the freezing process and propose the possibility of using it as a new screen material. We also describe a method for implementing an omni-directional ProCam system that reproduces the freezing phenomenon at room temperature and enables interactive projection onto the frozen soap film. We present a prototype-based application and discuss its technological potential and future research directions.

CCS Concepts

• Human-centered computing → Displays and imagers.

Keywords

projector camera system, frozen soap bubble, omni-directional display, shape changing display

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1 Introduction

Soap bubbles have long fascinated researchers and artists due to their unique optical and physical properties, including interference patterns, flexibility, and transience. In science and engineering, many studies have explored[2] and utilized this fascinating phenomenon[3]. In the field of Human-Computer Interaction (HCI), various technological attempts have been made to apply the unique beauty of the soap film as a display. However, since soap film is highly transparent, direct image projection is challenging as light passes through it rather than being diffused. For example, direct image projection onto the transparent soap film or a soap bubble has been achieved by creating surface irregularities through ultrasonic irradiation[7] or by trapping smoke inside the soap bubble[6].

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In this research, we further explore the possibilities of this beautiful soap film as a screen material, and propose a novel approach to screen opacification by “freezing the soap film”.

2 Characteristics of Soap Bubble Membrane Freezing Phenomenon and Its Potential

When exposed to temperatures below its freezing point, the soap bubble membrane may freeze (Figure 1(A-D)). As shown in Figure 1(D), the frozen areas of the soap film exhibit significantly reduced transparency compared to the unfrozen areas. In this research we focus on this transparency change of the soap film due to freezing as an element that enables direct image projection onto the soap film and also as an element that enables dynamic switching of the transparency of screens such as PDLC material. In the early phase of the freezing process of soap films, small localized frozen areas (ice flakes) appear somewhere on the film (Figure 1(B)). This is related to ‘ice nuclei,’ which form when airborne dust or frost particles adhere to the supercooled film or when frost from the ground is lifted by convection currents[1]. Then, the number and area of the ice flakes on the film gradually increase, and eventually, they cover the entire surface of the film, leading to the complete freezing of the film (Figure 1(D)). This type of soap film freezing is a visually striking natural phenomenon, particularly observed in cold climate regions. In this research, we focused on the characteristic properties of soap films during this freezing process that can be applied to screens.

First, a completely frozen soap film screen has a unique crystalline pattern formed by bonding of ice fragments of various shapes and sizes (Figure 1(D)). Therefore, the opacity of the membrane is characteristically non-uniform. A screen with such non-uniformity may not be suitable for general purposes such as viewing images or reading text. However, this non-uniformity is created by a natural phenomenon with unique randomness and beautiful visual characteristics, thus we focus on the possibility of unique image expression by combining this pattern and projected images.

In addition, the thin film (several nanometers) has shape and size variability from flat to spherical, and can easily change its state such as freezing, thawing, and breaking with little thermal energy. We believe that these properties also have potential for use as shape-changing displays and ephemeral screens. Furthermore, in addition to direct contact with the user, we believe that it also has the potential for interaction with non-contact elements such as temperature.

Second, frozen areas (ice flakes), which are locally formed and change dynamically during the freezing process, create both areas of high transparency (unfrozen areas) and areas of low transparency (frozen areas/ice flakes) on the same soap film. This is a unique and interesting property compared to conventional screen materials. Furthermore, the number, shape, and position of these ice shards are greatly affected by the composition of the bubble solution, temperature conditions and other factors, and they exhibit a variety of movements.

For example, if the temperature at the bottom of the soap film is higher than that at the top, small frost particles attached to the ground may be carried to the top of the film by convection currents. Then, ice flakes, which are formed from those small frost particles, grow and move around on the membrane, resulting in a

unique and beautiful phenomenon (snow globe phenomenon). On the other hand, if the temperature at the ground surface is lower than that at the top, freezing begins at the ground and gradually expands upward (bottom-up freezing). In this case, depending on the conditions, a beautiful dendritic pattern can be observed in accordance with the growth of ice fragments. In this research, we also focus on the possibility of applying this technology to new dynamic and interactive video expression by combining it with projector-camera (ProCam) technology, which enables real-time tracking and projection of ice fragments. We will also focus on the possibility of controlling technology that can create various conditions that affect the movement of ice flakes on the membrane, which changes dynamically during the freezing process.

In this paper, to explore the possibility of applying these unique properties of frozen soap films to new visual expressions, we first propose a new projector camera system that can easily reproduce the freezing phenomenon of soap films and can also dynamically project images onto the frozen film, and its implementation. Furthermore, we describe the applications implemented using the proposed system and discuss the possibility of future expansion.

3 RELATED WORK

Bubble Cosmos[6] utilizes trapped smoke as a medium for image projection onto floating soap bubbles. The phenomenon of the trapped smoke diffusing into the surroundings when the soap bubbles pop is also combined with the video projection. Bubble Cloud[4] is a shape-changeable display using a cluster of small soap bubbles, which enables projection of images without smoke trapping, and also proposes a technology to make images clearer by ultrasonic irradiation. Next, SensaBubble[8] generates a soap film filled with fragrant smoke and projects images on the surface of the bubble, attempting to present olfactory information that remains in place after it is broken, in addition to visual information on the soap film. Poppable Display[7] projects images onto a transparent soap film by applying ultrasonic waves to the film from the outside to create an uneven surface. FugaciousFilm[5], another ephemeral display, focuses on the fragility of soap film and proposes an interaction in which the fragile display is carefully manipulated.

As described above, previous research has attempted to utilize the unique beauty and fragility of soap film in displays through various methods. Our approach, focusing on the phase transition of soap film via freezing, differs fundamentally from previous methods and expands its potential applications, especially in dynamic and interactive displays.

4 Prototyping

The prototype and hardware configuration implemented in this study are shown in Figure 2(left). We aimed to inflate the soap film from a flat surface to a dome or sphere shape, and cool it to a frozen state at room temperature. In particular, to avoid blocking the user’s direct interaction with the soap film, such as “observing the soap film from a close distance” or “touching it by hands,” an open-type projection system (Figure 2(A)) was realized without covering the soap film with walls. The following is a detailed description of the prototype implementation we developed.

4.1 Soap film cooling mechanism

In order to freeze a grounded dome-shaped soap film, it is necessary to simultaneously cool the grounded part of the film in addition to cooling the air outside and inside the dome. In a room environment at room temperature, the outside (top) of the membrane is filled with warm air, so even if only the inside air or the ground is cooled, it is difficult to completely freeze the entire dome because the air outside the membrane immediately warms the inside air and the ground. Therefore, in this implementation, we attempted to replace the warm air around the soap film with cold air by sending cold air upward from right next to the base where the soap film contacts the ground (Figure2(left)). To achieve this, cold air from the freezer was drawn in using a fan, straightened through internal channels, and blown out around the bubble's base at a controlled rate. Similarly, for cooling the air inside the soap film, a built-in small pump can be used to suck up the cold air, compress it, and feed it into the film. Most of the main body of the prototype is embedded in a 162L freezer (JCMCC-162) that can be cooled to a maximum of -60°C , and cooled air can be blown directly into the inside and around the soap film at any airflow rate. In addition, for environmental impact and safety of the experience, we did not use liquefied gases such as carbon dioxide or nitrogen for cooling.

Next, the base of the soap film was circular with 110 mm diameter for this implementation, and the height of the base was adjusted so that there would be a distance of approximately 10mm between the soap film surface and the top of the fisheye lens head to avoid direct contact. The soap film can be inflated from a flat surface to a dome shape with a height of about 10 mm in about 5seconds after the pump is activated. To actually stretch the soap film, the user only needs to lightly scratch the entire base with a sponge that has absorbed the bubble solution.

4.2 Image Processing for Real-time Projection

For projecting and capturing images on the soap film, an ultra-wide-angle projector-camera system is embedded inside the dome of the soap film, making it possible to project and capture images of the whole area with only one ProCam, even when the film is inflated three-dimensionally and in a variety of sizes. The embedded ProCam mechanism consists of a small laser projector (LB-UH6CB) and a small infrared camera (FLIR FFY-U3-16S2M) coaxially connected to a fish-eye lens (Opteka 0.3XHD) via a beam splitter (EdmundOptics NIR20R/80T). In addition, the 28 infrared(940nm) LEDs light sources (OSI5LAS1C1A) were placed circumferentially around the fisheye lens(Figure2(B)), and a visible light cut filter (FUJIFILM IR-80) was attached to the camera.

Next, in order to realize image projection in accordance with the dynamically changing pieces of ice on the soap film, an application development platform was built on Unity that enables real-time(over 60fps) video capturing of the entire soap dome using the built-in camera and the generation of effect images using these images as input. These processes are performed by a shader program that uses IR images (Figure2(D)) taken from inside the soap film as input textures, enabling real-time synthesis of video effects combined with various image and video data. The generated results can be immediately projected from a coaxial projector to the entire surroundings(Figure1(H)).

The actual continuous projection onto the soap film from the beginning to the complete freezing of the film is shown in Figure1(E-H). The soap solution used was a 1:5:5 mixture of dishwashing detergent, PVC (polyvinyl alcohol), and water, based on a general recipe for soap solution.

5 APPLICATION

The freezing of soap film, explored in this research as a novel screen material, is a visually striking natural phenomenon. However, it is not easy to reproduce it outside of cold regions, and many people have never actually observed it up close and carefully. The proposed system, which can reproduce the freezing of soap film in a room at room temperature, will provide more people with various opportunities to experience this natural phenomenon and to become interested in the various physical phenomena behind this beautiful phenomenon.

In addition, the experience of combining the frozen soap film and interactive video projection provided by the proposed system includes a variety of preparatory steps such as "preparing the soap solution," "stretching the film," and "inflating the bubble" as well. These experiences are also enjoyable and have a greater impact on the final result than the expansion technology and projection experience described below, so we position the user's own experience of them as one of the important interactive elements of this experience.

5.1 Immersive experience enhancement through multiple projection onto the surrounding environment

The proposed system can visually enhance the beautiful frozen phenomenon of the soap film by projecting images onto it, but images projected onto the transparent (unfrozen) area on the film pass through the film. This multi-layered projection capability enables, for example, the simultaneous projection of immersive images onto the user's surrounding environment (walls, ceiling, etc.) in addition to the projection of images onto the soap film. This would also make it possible to further extend the projected images on the soap film with ambient projected images in the user's surrounding environment. In addition, since the projected image on the surrounding environment is greatly magnified, the shape and movement of small, invisible ice fragments on the soap film can be easily observed.

5.2 Melting a frozen film

Soap films are extremely thin and can be thawed (re-transparent) with low thermal energy after freezing. This property allows the system/user to interact with the frozen soap film in a variety of ways. For example, in addition to direct contact with the user's hand or fingers, the frozen film can be intentionally heated by getting close to the user's hand or fingers, or blowing breath on the film to thaw it. Furthermore, the membrane can be easily thawed using heat sources such as projector lights or laser pointers. The user can visually enjoy the changes in the frozen state that occur as a result of these physical and tool actions through trial and error. It is also easy for the user to physically destroy the frozen soap bubbles directly with users' hands to remove them completely. This property could be used to create an ephemeral screen just before

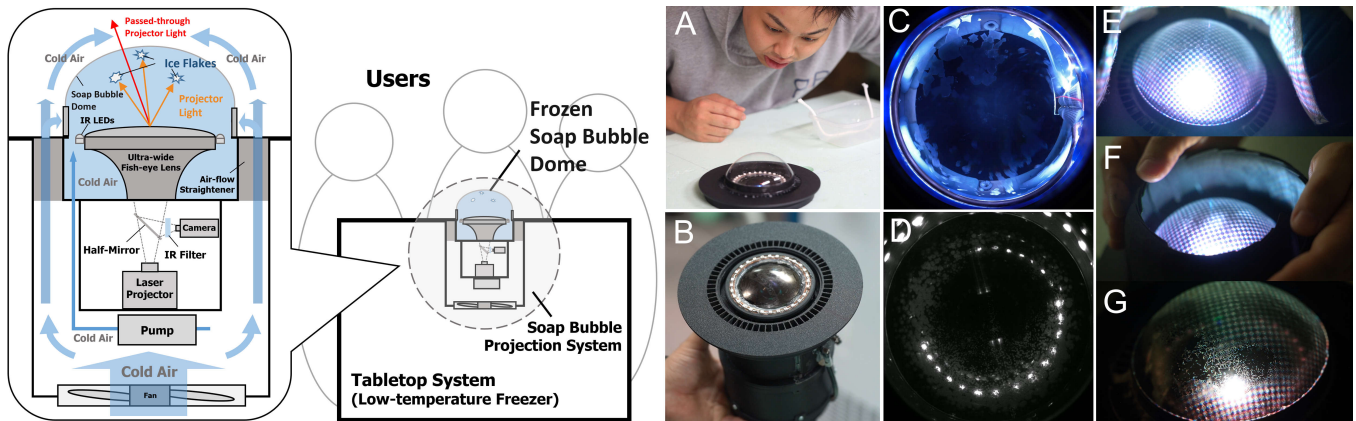


Figure 2: Hardware configuration (left) and operation of each part of the implemented prototype (A-G).

the projector lens, which can be created when necessary and then destroyed when finished, and thus disappear.

5.3 Extending the experience with hands and tools

Our implemented system is a fully open type table-top system, which allows users to interact with the soap film, such as by directly approaching or touching it. Therefore, users can have an extraordinary experience of freezing a soap bubble as if they were “warming” it with their hands while holding it gently with gloved hands shown in Fig2(E). (Note that the user wears winter gloves for this experience in order to avoid direct skin contact with cold air.) In this application, the cold air blown from the bottom of the table directly hits the user’s hand, allowing the user to directly control the direction and strength of the cold air by changing the position and angle of the hand and fingers. For example, if the user gently wraps the entire soap bubble, the freezing speed of the soap bubble is increased due to the air trapped inside the hand. Also, if the user opens his or her hand in one area, only that area will be thawed again, or the movement of the ice flakes will change due to changes in convection. These interactions allow the user to physically and directly interact with the soap film during the freezing process, which is difficult to achieve with natural freezing.

Similarly, interaction can be extended by using various tools. For example, the figure2(F) shows a soap film screen with 3D printed retractable vane parts. The vanes can trap the soap-dome inside and store cold air. The user can adjust the degree of opening of the blades by hand or motor, and by opening and closing the blades in combination with the content animation, it is possible to create an expression like a beautiful flower opening with soap bubbles inside.

5.4 VR experience inside a frozen soap dome

This VR application allows the user to freely look around the freezing soap film from inside the soap bubble dome. Although it is difficult to actually freeze a huge dome-shaped balloon that can accommodate people, it is possible to reproduce the scene virtually by using VR goggles to play back the images captured by the

camera inside the dome(Figure2(C)). In this application, a high-resolution visible light camera (DC-GH6) and its corresponding ultra-wide-angle lens (Entaniya HAL250) are used instead of an infrared camera.

6 Future Work

In future work, further study is needed to improve the durability of the soap film as it relates to the duration of the experience. In particular, we found that the part of the film that was thawed again after being frozen had small holes on the surface due to the drying of the ingredients in the film, which made it easy to tear(Figure2(G)). We will continue to investigate the optimal liquid mixture for freezing.

In the current prototype, precise systematic control over the freezing conditions has not yet been achieved. In the future, we aim to develop a programmable freezing process, for example, by controlling convection strength through temperature regulation of the film’s base to generate snow globes or selective bottom-up freezing.

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